From Wing Lab to STL

MIE 446 | Notebook steps | Code-to-Print release v1.3.1

Start here: Your fit coupon has already been made. Design and analyse your wing in Wing Lab first. This notebook converts your exported design into CAD and print files; its quick geometry report does not repeat the structural analysis.

Before the notebook: design your wing

In Wing Lab, choose your wing, add the load, check the root support and press **Solve**. In your saved Colab copy click **+ Text**, title it **Design basis**, and write: Why this wing? What load (value, unit, location and direction), and why? What is your target, such as maximum tip displacement, and does the result meet it? Add a screenshot. Ask the instructor if you need help choosing a load or target. This is a text cell you add; it is not a field in the existing form. Keep a **Save JSON** backup, then **Export to notebook**. Each printed part must fit the 300 x 300 x 300 mm printer. Loads are virtual only.

1. Start the design tool

Open the current notebook in Colab. Select **File > Save a copy in Drive** and work in that copy. Connect the runtime and run the setup cell under **1. Start the design tool**. Wait for **Design tool ready**. A GPU runtime is not required.

2. Complete STUDENT INPUT FORM

Set **DESIGN_SOURCE = Wing Lab JSON** and **WORKFLOW_STAGE = Final Wing**. Enter the actual **TEAM_ID**, **TEAM_MEMBERS** and **REVISION**. Leave **WING_LAB_JSON_FILE** blank for the upload chooser. Keep **PRINTER_X_MM / Y_MM / Z_MM = 300 / 300 / 300**. Geometry fields in section B do not replace the uploaded design.

External rods: enter your existing physical test record in section C: **COUPON_CONFIRMED**, **COUPON_REVISION**, **MEASURED_ROD_DIAMETER_MM**, **SELECTED_RADIAL_CLEARANCE_MM**, **COUPON_FIT_RESULT**, **COUPON_TESTER** and **COUPON_DATE** (YYYY-MM-DD). Use a final-wing revision different from the coupon revision. Do not invent a measurement. Without external rods, the notebook marks the coupon stage not applicable.

Reuse coupon evidence for the same rod, clearance and print process; have staff review any changed interface.

Before Step 3: complete section D's airfoil reason, comparison tip chord, predicted area/AR changes and prediction explanation. In section E declare AI use. If AI materially affected the work, complete its verification record. Run the form.

2A. Import the Wing Lab design

Run **Import the Wing Lab design (only for Wing Lab JSON mode)**. Click the upload chooser and select exactly one JSON from **Export to notebook**. Check the imported summary against Wing Lab: airfoil, span, chords, modules and components. The form values alone do not upload a file.

Keep the design in one place. Make geometry changes in Wing Lab, solve again and export a new transfer JSON. Rerun the form and import, then all later cells. The import must match the wing you intend to print.

Build, download and print

Continue in the same saved notebook, from top to bottom.

3. Preview and understand the design

Run **Run the automatic 2D analysis**. Compare the airfoil/planform, projected dimensions, rib layout, area and taper with your design. Check one value by hand and review the tip-chord comparison. Complete **RESULT_INTERPRETATION** and **PRINT_DEFECT_CODE_CANNOT_DETECT** in section D. After changing the form, rerun Step 2, the upload cell, and Step 3 before continuing.

4. Use the completed coupon record

Run **Run the coupon stage** in **Final Wing** mode. With external rods, wait for **Coupon record accepted**. With no rods, the notebook reports that a rod-fit coupon is not applicable. You are recording the test already performed, not starting another Coupon Only run.

5. Build and verify the 3D wing

Run **Build and inspect the 3D model**. Wait for the actual CAD build and validation; the notebook displays the printable modules. Check the geometry, module count, actual CAD bounding boxes and mass. Passing the earlier 2D preview is not enough. The final plastic mass limit is **300 g**.

One piece or modules? One module means one continuous semi-wing. Every actual CAD part must fit the configured usable printer envelope. The notebook checks axis permutations, not arbitrary-angle packing. A straight 450 mm part fails the 300 x 300 x 300 mm check; return to Wing Lab and choose a fitting modular design. Never enlarge the printer fields to bypass the check.

6. Create and download the final ZIP

Run **Create and download the final ZIP**. The notebook checks the required written responses and AI declaration. Wait for **Submission package ready**, then save the ZIP locally. It contains STEP/STL/3MF geometry, parameters, calculations, validation, team/revision records and file hashes. Preserve the original ZIP and extract a working copy.

7. Open the STL in OrcaSlicer and build

Import the STL for **every module**. Select the assigned **K2 Pro / 0.4 mm / staff-approved PLA Pro profile**. Confirm **millimetres and 100% scale**, match the CAD dimensions, allow brim/support space, and inspect every layer. Check estimated mass and time. Obtain staff print approval, follow the assigned printer procedure and supervise the first layers. A passed CAD check is not print approval.

Keep: editable Wing Lab JSON, transfer JSON, saved notebook, original final ZIP, slicer project and build/assembly photographs. Record actual dimensions, mass and fit. Use the same team and revision identifiers. Keep the team wing unloaded and undamaged.

Detailed printer operation: course OrcaSlicer / USB guide. Canvas carries the assigned profile, reservations, submission and deadlines.